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Governance Effects and the Role of On and Off-Chain Vote Divergence in Decentralized Autonomous Organizations (DAOs)

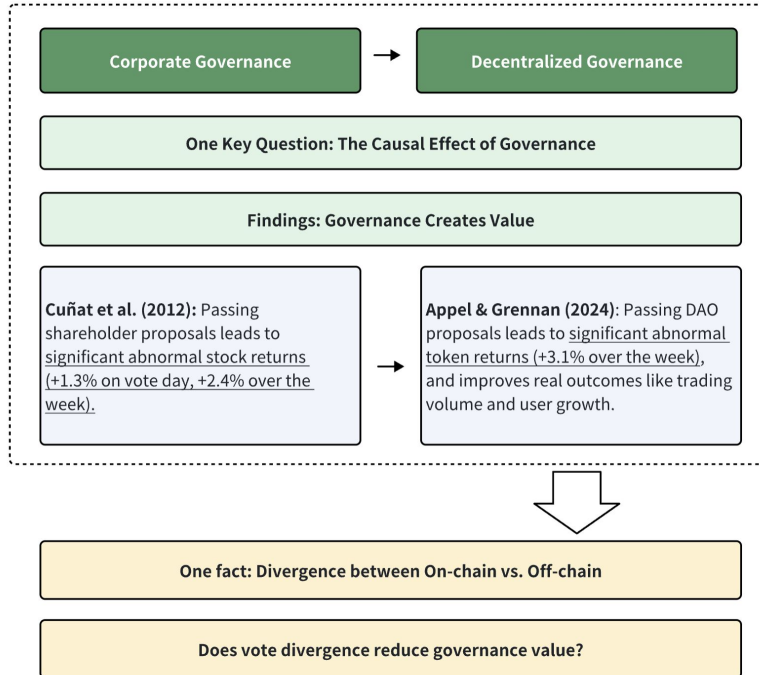
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Literature Background and Research Question

From Corporate Governance to DAO Governance: Does Vote Divergence Matter?



Governance and Value

Both corporate governance (Cuñat et al., 2012) and DAO governance (Appel & Grennan, 2024) show that passing proposals creates positive abnormal returns.

A New Challenge in DAO Governance

- On-chain voting is costly and dominated by large holders.
- Off-chain voting tends to attract broader community participation.

Research Question

Does divergence between on-chain and off-chain outcomes weaken the value that governance creates, given that off-chain voting reflects broader community preferences?



Motivation

Evidence of Systematic Preference Gaps between On and Off-Chain in DAO Governance

Figure 1

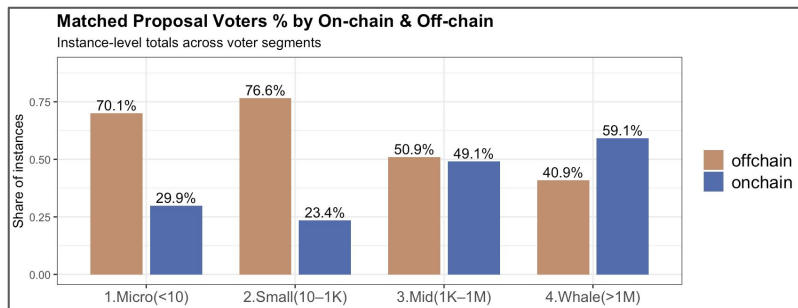
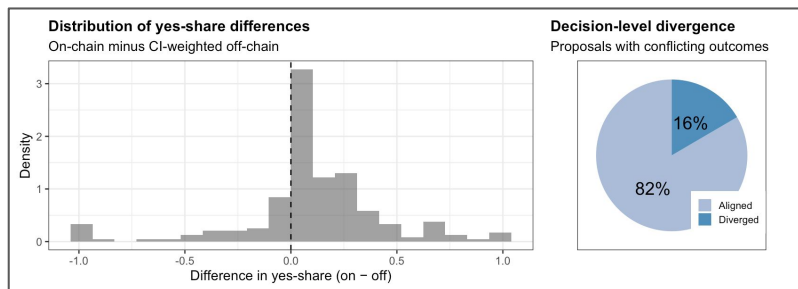


Figure 2



Why Study Vote Divergence?

Participation Structure

Small and retail voters participate far more in off-chain voting, while large holders dominate on-chain voting (Figure 1).

Preference Divergence

On-chain and off-chain yes-share often differ substantially for the same proposals, and 16% of proposals even yield conflicting pass/fail outcomes across the two systems (Figures 2–3).

Motivating Question

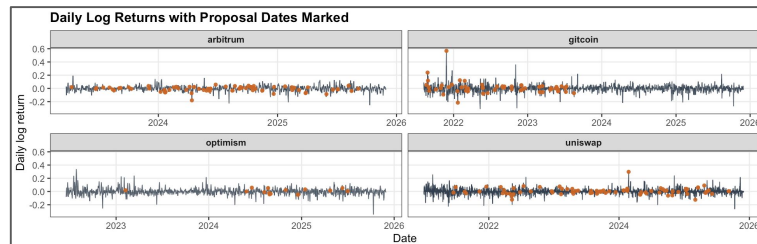
How do such divergences affect the governance effects that DAOs create?



Data and Methods

Identifying Governance Effects and the Role of Divergence

Figure 4



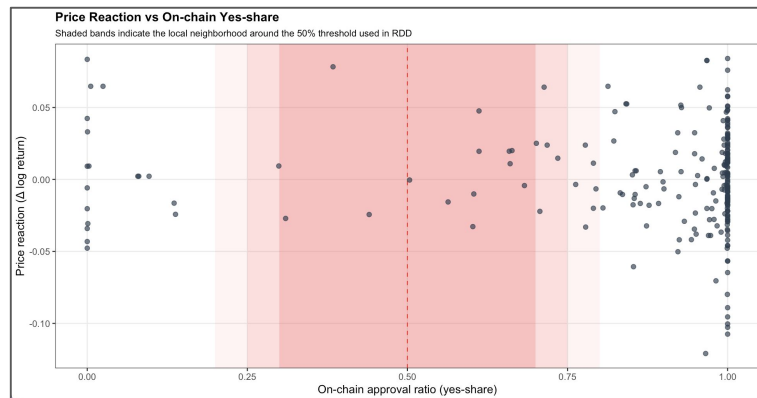
Data

From 2021-01 to 2025-11 from Dune platform.

Outcome (Y)

Cryptocurrency log returns around proposal events (short-window price reaction, as figure 4)

Figure 5



Regression Discontinuity Design (RDD)

- Omitted Variable Bias: DAOs with strong governance features may also differ in community engagement, developer quality, market positioning, etc.
- Quasi-experiment: 50% approval threshold as quasi-experiment to identify causal effect of passing (as figure 5).



Results

Positive Governance Effects Exist, but Divergence Weakens Governance Value

Table 1

RDD Estimates under Different Bandwidths				
	(1) h = 0.20	(2) h = 0.25	(3) h = 0.30	(4) All
Constant	-0.057 (0.038)	-0.055 (0.038)	-0.084* (0.040)	0.011 (0.018)
Pass	0.113 (0.059)	0.119* (0.057)	0.147** (0.057)	0.022 (0.028)
Governance gap	0.608** (0.165)	0.603*** (0.189)	0.663*** (0.214)	0.004 (0.039)
Pass × gap	-0.658*** (0.177)	-0.687*** (0.195)	-0.732*** (0.218)	-0.031 (0.043)
Off-chain pass	-0.303*** (0.067)	-0.302*** (0.079)	-0.314*** (0.090)	-0.028 (0.028)
Pass × off_pass	0.249** (0.070)	0.246** (0.082)	0.274*** (0.092)	0.009 (0.031)
Running var.	0.081 (0.120)	0.091 (0.090)	-0.041 (0.071)	-0.025 (0.023)
R-squared	0.837	0.696	0.485	0.026

* h denotes the bandwidth ($|\text{running_var}| \leq h$); smaller h restricts the sample closer to the 50% approval threshold.
† 'All' uses the entire sample without any bandwidth restriction.

Key Variables

- **Pass**: effect of passing a proposal
- **Pass × gap (main)**: how divergence modifies that effect

Positive Governance Effect

Governance in DAOs increases returns.

Negative Divergence Effect

On-off vote divergence weakens the positive governance effect of passing a proposal. Coef = -0.658: a 0.01 increase in the yes-share gap reduces the governance effect by around 0.65.8%

Robustness Across Bandwidths

Four RDD models using bandwidths 0.20, 0.25, 0.30, and All yield consistent patterns.



Conclusion and Future Research Directions

Data expansion and research extensions in heterogeneity ...

Conclusion

- Positive governance effects exist.
- Divergence weakens governance value.
 - Off-chain voting captures valuable information about broader community preferences.

Future Research Directions

- **Data expansion:** More DAOs and proposals, longer sample period, and richer outcome measures (e.g., trading volume, user activity, protocol usage).
- **Research extensions:** Explore heterogeneity across DAOs, voter segments, and proposal types.